# COMP 352
# Data Structures and Algorithms
# Course Outline – Winter 2026

GINA CODY SCHOOL OF ENGINEERING AND COMPUTER SCIENCE

| | |
|---|---|
| **Instructor** | **Section S:** Dr. Hamed Jafarpour - Office: H961-15<br>E-mail: hamed.jafarpour@concordia.ca (The email subject must contain "COMP 352")<br>Lecture:<br>Monday – Wednesday: 1:15 PM - 2:30 PM at H-937<br>Office Hours: 1) Hybrid / Mondays (10:00 AM - 11:30 AM)<br>2) Hybrid / Thursdays (1:30 PM - 3:00 PM)<br>Check Zoom link on Moodle |
| **Instructor & Coordinator** | **Section X:** Dr. Aiman Hanna - Office: ER_1103<br>E-mail: contact@AimanHanna.com Tel: (514) 848-2424 ext. 7878<br>Lecture:<br>Thursday 5:45 PM - 8:15 PM at H-535<br>Office Hours:<br>1) Thursday 3:30 PM – 4:30 PM; 2) By appointment anytime. |
| **Fastest way to communicate:**<br>**Through the *Replied* App at the *COMP352W26* group** | |

## Tutorials & PODs:

Tutors, PODs and markers can be contacted on Replied at ***COMP352W26*** group.

**Tutorials:**

| Tutorial | Tutor | Replied ID |
|---|---|---|
| **XA** Mo 4:15PM - 5:05PM at H-827 | **Ryan Atallah** | **@ryan_atallah** |
| **XB** - We 4:15PM - 5:05PM at H-847 | **Aya Al Haj** | **@aya** |
| **XC** - Mo 5:45PM - 6:35PM at H-847 | **Kassem Tawbi** | **@kassemtawbi** |
| **SA** - We 2:45PM - 3:35PM at H-823 | **Disha Padsala** | **@disha** |
| **SB** - Mo 2:45PM - 3:35PM at H-827 | **Ahish Mahesh** | **@ahish.mahesh** |
| **SC** We 2:45PM - 3:35PM at H-827 | **Shariq Anwar** | **@shariqanwar20** |

**PODs:**

| POD<br>Times and locations | POD Instructor | Replied ID |
|---|---|---|
| Monday 10:00 AM - 11:00 AM @ H-921 | Husein Hijazi | @huseinhijazi |
| Wednesday: 5:00 PM - 6:00 PM @ H-819 | Osama Iskandarani | @_isk |
| Monday 6:00 PM - 7:00 PM @ H-921 | Rushil Tejashkumar | @rushil |

**Markers:**

| Marker | Replied ID |
|---|---|
| Mario Chahoud | @mariochahoud |
| Tarek Hamze | @tarekhamze |
| Parmeet Singh | @parmeetsingh |
| Sneha | @sneha17 |
| Saranraj Sivakumar | @saranraj_s |
| Ricardo Chahine | @kassemtawbi |
| Dev Patel | @devpatel15 |
| Manush Shah | @manush1811 |
| Jaiminkumar Mayani | @jaiminmayani |
| Taksh Rana | @taksh |
| Dhruvil Shah | @dhruvilshah00 |
| Shashwat Shah | @shashwat_shah |

**Pre-Requisites:** COMP 232 or COEN 231; COMP 249 or BCEE 231.

**Calendar Course Description:** Abstract data types: stacks and queues, trees, priority queues, dictionaries. Data structures: arrays, linked lists, heaps, hash tables, search trees. Design and analysis of algorithms: asymptotic notation, recursive algorithms, searching and sorting, tree traversal, graph algorithms.

**Algorithms and AI:** This course incorporates foundational elements of Artificial Intelligence by highlighting how core algorithms and data structures are used in real AI systems. You will be exploring how the covered structures (stacks, queues, graphs, trees, … ) and algorithms enable intelligent behavior in areas such as search, planning, natural language processing, and reasoning. Through selected written/programming assignments and case-based analysis, the course directly connects classical algorithmic thinking to modern AI applications. No prior knowledge of AI is required, but you will gain early exposure to the algorithmic building blocks that power today's intelligent technologies.

## Course Materials

## Textbook

M.T. Goodrich, R. Tamassia, Michael H. Goldwasser. Data Structures and Algorithms in Java, 6th edition. John Wiley & Sons, 2014. ISBN 978-1-118-77133-4. (Note: 5th edition is ok.)

The book is available at the bookstore or can be rented as eTextbook.

Textbook/eTextbook U R L : http://ca.wiley.com/WileyCDA/WileyTitle/productCd-EHEP002900.html

From the textbook we shall study these sections: 3.1 to 3.4, 4.1 to 4.3, 5.1 to 5.6, 6.1 to 6.2, 7.1 to 7.3, 8.1 to 8.4, 9.1 to 9.4, 10.1 to 10.3, 11.1 to 11.3, 12.1 to 12.4, and 14.1 to 14.7.
**You also need to study chapters 1 and 2 that will not be covered in provided lectures**.

## Grading Scheme

The table illustrates the components of the course and their corresponding weights (%).

| **Component** | **%** |
|---|---|
| Assignments | 10% |
| Midterm | 30% |
| Final Exam | 60% |

There is **no fixed**, a priori relationship between the numerical percentage and the final letter grades for this course. To pass the course, you **must attempt all assignments**, pass the midterm, and pass the final exam. Usually a score of 50% is required for passing the exams. There are no make-ups/alternates for missed exams or assignments. The instructors reserve the right to modify the grading scheme under exceptional circumstances to accommodate certain situations (e.g., medical).
In the event of extraordinary circumstances beyond the University's control, the content and/or evaluation scheme in this course is subject to change.

**The Midterm Exam, for all sections,** is scheduled for **Saturday, March 14, starting at 2:30 PM.** **There is no substitution for a missed exam. Your instructor will indicate the exact location of the midterm.**

## Details of the Course Components

1. **Assignments:** There will be 3 written assignments and 3 programming assignments (each weight depends on the difficulty of the assignment), all of which will be available online. These assignments play a major role in your learning of the various topics covered in this course. **You must attempt all assignments to pass the course.** Assignments must be submitted by the **due date** via Moodle. There will be a 20% per day **penalty** for late submissions. No assignment will be accepted after two days of delay. More detailed information on assignments is provided below. Please read them carefully!

## Important Notes

i) Assignments will consist of *written assignments* and *programming assignments*. Each student must **independently** and **separately** prepare and submit the written assignments **(no groups are allowed).** For the programming assignments, you are allowed to work in a group (maximum 2 students).

ii) **For the written assignments:**
- Please read carefully: For the written assignments, only one or two questions will be marked at random; however, you MUST submit the answers to all questions; otherwise, marks will be deducted for each missing answer. Missing the questions that are randomly selected for marking will result in a zero mark of the assignment, even if all other questions are submitted.

iii) **For the programing assignments:**
- In case you are working in a group, ONLY ONE copy per group is to be submitted; i.e. do **not** submit 2 separate copies. You have to submit the complete source code **and** the compiled files, which can be executed without changes. If this is violated, you will get a zero mark for these parts of the assignments.

- You need to use Java version 8.0 or later. You will be using the same computing facilities and the same computer account you used in previous courses (e.g., Comp 249). If you do not have a computer account, you can obtain it from the help desk at H-960 or EV_07.182. This account will give you access to the laboratories. For more information on CSE Computer accounts please visit the following website: http://www.encs.concordia.ca/helpdesk/access.html. If you have your own computer and prefer to use it, you may do so, but be aware that your programs must compile and run with Java 8.0, or later version, at the Concordia laboratories.
- **IMPORTANT (Please read very carefully):** Additionally, which is very important, for the programming assignments, a demo will take place with the markers afterwards. Markers will inform you about the details of demo time and how to book a time slot for your demo. If working in a group, both members must be present during demo time. Different marks may be assigned to teammates based on this demo. **Now, please read very carefully:**
    a. **If you fail to demo, a zero mark is assigned regardless of your submission.**
    b. **If you book a demo time, and do not show up, for whatever reason, you will be allowed to reschedule a second demo but a penalty of 50% will be applied. There will be no exceptions to this rule; so please do not ask!**
    c. **Failing to demo at the second appointment will result in zero marks and no more chances will be given under any conditions.**

iv) **Submission format:** All assignment-related submissions must be adequately archived in a ZIP file using your ID(s) and last name(s) as file name. The submission itself must also contain your name(s) and student ID(s). Use your "official" name only - no abbreviations or nick names; capitalize the usual "last" name. Inappropriate submissions will be heavily penalized. Only electronic submissions will be accepted. Students will have to submit their assignments (one copy per group for the Programming Part) using Moodle. Assignments must be submitted in the right folder of the assignments. **Assignments uploaded to an incorrect folder will not be marked and result in a zero mark. No resubmissions will be allowed.**

2. **Midterm and Final exams**: We will have a midterm and a final exam in this course, which will take place in-person at Concordia campus. The formats of these two exams will be similar and include multiple choice questions (with negative marks for wrong answers), questions with detailed answers, and True/False questions requiring a short justification. The final exam will be identical for all sections. **You must pass the midterm and pass the final exam in order to pass the course.**

## Tentative Course Schedule

The schedule is **tentative** and might change anytime.

| Chapter | Topic | Week |
|---|---|---|
| - | Introduction | 1 |
| 4 | Algorithm Analysis (4.1 to 4.3) | 1 & 2 |
| 5 & 15 | Recursion (5.1 to 5.6) | 2 |
| 3 & 6 | Arrays, Stacks and Queues (3.1, 6.1 and 6.2) | 3 |
| 3 & 7 | Lists (3.2 to 3.4, and 7.1 to 7.3) | 4 |
| 8 | Trees (8.1 to 8.4) | 5 & 6 |
| 9 | Priority Queues and Heaps (9.1 to 9.4) | 6 & 7 |
| 10 | Maps and Hash Tables (10.1 to 10.3) | 8 & 9 |
| 11 & 12 | Search Trees (11.1 to 11.3) & Sorting (12.1 to 12.4) | 9 & 10 |
| 14 | Graphs (14.1 to 14.6) and 14.7 (if time allows) | 10, 11 & 12 |

## Graduate Attributes

As part of either the Computer Science or Software Engineering program curriculum, the content of this course includes material and exercises related to the teaching and evaluation of graduate attributes. Graduate attributes are skills that have been identified by the Canadian Engineering Accreditation Board (CEAB) and the Canadian Information Processing Society (CIPS) as being central to the formation of engineers, computer scientists and information technology professionals. As such, the accreditation criteria for the Software Engineering and Computer Science programs dictate that graduate attributes are taught and evaluated as part of the courses. The following is the list of graduate attributes covered in this course, along with a description of how these attributes are incorporated in the course:

- **Problem analysis:** *Ability to use appropriate knowledge and skills to identify, analyze, and solve complex engineering problems in order to reach substantiated conclusions.* Analyze problems and determine their constraints in order to make a choice as to what data structures and algorithms to use for their implementation.

- **Design:** *Ability to design solutions for complex, open-ended engineering problems and to design systems, components or processes that meet specified needs with appropriate attention to health and safety risks, applicable standards, and economic, environmental, cultural and societal considerations.* Use and compose appropriate data structures and algorithms to solve a variety of problems.

## Course Learning Outcome (CLOs)

- **Problem analysis:**
  Analyze and state model limitations and elements of uncertainty. Formulate and calculate qualitative and quantitative qualities of the problems inputs and outputs. Estimate computational complexity. Evaluate and pick the most appropriate approach based on relevant criteria.
- **Design:**
  - Critique/evaluate many possible diverse solutions and use techniques to evaluate different solutions with sound arguments related to the problems requirements and constraints. Demonstrate thinking outside the box to create innovative solutions.
  - Develop a system architecture adapted to the systems application context and its requirements and constraints. Development and specification of internal and external software interfaces at different modularity levels. Describe a solution that presents enough details for implementation.
  - Write code according to design. Validate implemented systems against system requirements, specifications and constraints, as well as interface specifications.
- **One credit** represents, for an average student, a minimum of 45 hours of workload spread across the various academic activities (Source: Article 16.1.2 of the Undergraduate Calendar.) For an average student, this suggests a minimum of 135 hours of workload for a 3-credit course, including the time spent in lectures, tutorials, laboratories, examinations, and personal and team work.

## Web pages and other Resources

### The fastest way to communicate, including communications with the TAs and markers, is through the replied app at *COMP352W26* group.

## Web Pages

Many resources for the course (slides, assignments, example programs, ...) will be available online through Moodle; (available through the MyConcordia portal www.myconcordia.ca), or through the instructor's website. Your instructor will inform you of the exact link to access the materials.

For Sections S (Dr. Jafarpour), please use the Moodle Web site available through the MyConcordia portal www.myconcordia.ca.

For Sections X (Dr. Hanna), the webpage for the course is: **www.AimanHanna.com** (follow Concordia links afterwards). Other material will be available on Moodle as well; so you will need to consult both locations. The web pages will contain announcements related to the class, pointers to documents, your written and programming assignments, etc.

Additionally, the fastest way to communicate is through the replied app at the ***COMP352W26*** group**.** General questions for the course, and technical questions, should be communicated in that group only. For private messages, please create a private group and add ***@dr.hanna*** to it.

Further, a mailing list will be established for the course. You ***will need to subscribe*** to the mailing list. You can subscribe at the following link:
https://mailman.encs.concordia.ca/mailman/listinfo/comp352-W26.

Finally, the faculty web pages have a wealth of information pertaining to our computer systems and software, which include simple user guides, and answers to many standard questions. You should explore these help pages. Begin your exploration from the URL: http://www.encs.concordia.ca/helpdesk/faq/faq.php

# Important Lecture Guidelines (Section X)

Laptops are **STRICTLY PROHIBITED** during the lectures. Other communications devices, such as cellular phones, communication watches, and text/video messaging devices, tablets, pads, and similar devices are also **STRICTLY PROHIBITED**. The usage of any of these materials during the class will result in you being asked to immediately leave the class.

## Health and Safety Guidelines

All health and safety rules specific to this course can be found in the lab manual. General health and safety instructions and available health and safety trainings can be found at: Safety Programs - Concordia University (https://www.concordia.ca/campus-life/safety/general-safety.html).

## Plagiarism

The most common offense under the Academic Code of Conduct is plagiarism which the Code defines as *"the presentation of the work of another person as ones own or without proper acknowledgement."*
This could be:

- material copied word for word from books, journals, internet sites, professors course notes, etc.
- material that is paraphrased but closely resembles the original source.
- the work of a fellow student, for example, an answer on a quiz, data for a lab report, a paper or assignment completed by another student.
- a solution or Java code purchased through one of the many available sources.
- ➔ You must also notice that the submission of AI-generated material is strictly prohibited. The

submission of such contents is considered as direct plagiarism and violates the University Code of Conduct. Any such submissions, partially or fully, will result is an immediate plagiarism case being submitted to the University.

Plagiarism does not refer to words alone; it can also refer to copying images, graphs, tables, and ideas. Presentation is not limited to written work. It also includes oral presentations, computer assignments and artistic works. Finally, if you translate the work of another person into French or English and do not cite the source, this is also plagiarism.

In Simple Words:

**Do not copy, paraphrase or translate anything from anywhere without saying where you obtained it!**

## IP [VERY IMPORTANT: PLEASE READ CAREFULLY]:

Course contents belonging to the instructor and the TAs, including, but not limited to, lectures, course notes, and video recordings of classes/tutorials remain the intellectual property of the faculty member and the TAs. It must not be distributed, published or broadcasted, in any form or shape, in whole or in part, without an explicit written permission of the faculty member or the TA. It is also prohibited for students to use their own means of recording of any elements of a class, lecture, tutorial, etc., without explicit written permission of the instructor. Any unauthorized sharing of course content will constitute a breach of the Academic Code of Conduct and/or the Code of Rights and Responsibilities. As specified in the Policy on Intellectual Property, the University does not claim any ownership of, or interest, in any of such IPs; all university members retain copyright over their work.

In addition, please note the following important information concerning the assignments and the labs: As the assignments remain the sole property of the course instructor, you are not permitted to post the assignments or their solutions anywhere on the Internet either during the course or at any future point of time. Intellectual Property rights are reserved. Violation of such rules during the course, or at any future point, will result in both academic and legal actions.